

Software Requirements Specification (SRS

AI Financial Auditing Platform

Introduction .1

Purpose 1.1

This document describes the software requirements for the **AI Financial Auditing Platform**, a system that automates financial auditing processes using Artificial Intelligence and Machine Learning.

The platform aims to assist auditors, accountants, and regulatory bodies in detecting financial anomalies, ensuring regulatory compliance, and improving audit efficiency.

Scope 1.2

The system will provide

- Automated financial data analysis
- AI-based anomaly detection
- (Compliance verification (Tax & Accounting standards
- Risk scoring of financial transactions
- Audit report generation
- Dashboard for auditors and financial analysts

The platform targets

- Accounting firms

Financial auditors •

Enterprises •

Government regulators •

Tax authorities •

Definitions, Acronyms, Abbreviations 1.3

Meaning	Term
Artificial Intelligence	AI
Machine Learning	ML
Natural Language Processing	NLP
Optical Character Recognition	OCR
Record of all system actions	Audit Trail
Saudi Tax Authority	ZATCA
Value Added Tax	VAT

Intended Audience 1.4

:This document is intended for

Software engineers •

System architects •

Project managers •

Financial auditors •

Investors •

Regulatory authorities •

Overall Description .2

Product Perspective 2.1

:The system is a **cloud-based financial auditing platform** that integrates

Accounting data •

Financial transactions •

Invoice systems •

Tax compliance data •

AI models •

The platform processes financial records automatically and produces **audit insights and risk indicators**

Product Functions 2.2

:Main functions include

Upload financial data .1

AI financial analysis .2

Fraud detection .3

Tax compliance verification .4

Audit reporting .5

Dashboard visualization .6

Risk scoring .7

Document analysis .8

User Classes 2.3

Auditor

:Responsibilities

- Review audit results •
- Generate reports •
- Verify anomalies •

:Permissions

- Access financial analysis •
 - Generate audit reports •
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Financial Manager

:Responsibilities

- Upload financial records •
- Review compliance reports •

:Permissions

- View dashboards •
 - Upload documents •
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System Administrator

:Responsibilities

- Manage users •
- Configure AI models •

Monitor system activity •

:Permissions

Full system access •

Operating Environment 2.4

:The platform will operate on

:Frontend

Web Browser •

Mobile responsive UI •

:Backend

Python •

Django Framework •

:Database

PostgreSQL •

:AI Components

Python •

OpenAI models •

Machine learning libraries •

:Infrastructure

Cloud servers •

Docker containers •

Design Constraints 2.5

:Constraints include

- Compliance with financial regulations
- Data privacy laws
- High security requirements
- Scalability for large financial datasets

Assumptions and Dependencies 2.6

:Assumptions

- Users provide valid financial data
- Integration with accounting systems is available

:Dependencies

- AI models
- OCR services
- Cloud infrastructure

System Features .3

Financial Data Import 3.1

:The system shall allow users to upload financial data in formats such as

Excel •

CSV •

PDF •

Accounting system exports •

:Features

Data validation •

Data mapping •

Automatic classification •

AI-Based Financial Analysis 3.2

:The system shall analyze financial data using AI algorithms to detect

abnormal transactions •

unusual financial patterns •

accounting inconsistencies •

:Outputs include

anomaly reports •

financial insights •

risk indicators •

Fraud Detection 3.3

:The system shall identify potential fraud using

- pattern recognition

- anomaly detection

- behavioral analysis

:Examples

- duplicate invoices

- suspicious payments

- irregular transactions

Compliance Checking 3.4

:The system shall verify compliance with

- VAT regulations

- accounting standards

- tax requirements

:Examples

- incorrect tax calculations

- missing invoices

- regulatory violations

Audit Report Generation 3.5

:The system shall generate automated audit reports including

- financial risk summary

anomaly detection results •

compliance status •

:Report formats

PDF •

Excel •

Dashboard view •

Dashboard 3.6

:The system shall provide dashboards displaying

financial risk score •

anomalies detected •

compliance status •

audit history •

:Visualization

charts •

graphs •

indicators •

External Interface Requirements .4

User Interface 4.1

:The platform will provide

- web-based dashboards •
- interactive charts •
- data upload interface •

:Frameworks

- Django Templates •
- TailwindCSS •
- Alpine.js •

API Interface 4.2

:The platform will expose APIs for

- accounting software •
- financial systems •
- external audit tools •

Hardware Interface 4.3

.No specialized hardware required

:Users access system via

- computers •
 - tablets •
 - mobile devices •
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Non Functional Requirements .5

Performance 5.1

:The system shall

- process financial datasets up to 1 million records •
 - generate audit reports within 10 seconds •
 - support multiple concurrent users •
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Security 5.2

:Security requirements include

- encrypted data transmission •
 - role-based access control •
 - audit logs •
 - multi-factor authentication •
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Reliability 5.3

:The system shall ensure

- uptime 99.9% •
 - automatic backups •
 - fault tolerance •
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Scalability 5.4

:The system must scale to support

- large enterprises •
 - multi-company environments •
 - cloud deployment •
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Maintainability 5.5

:The system architecture should support

- modular AI components •
 - easy updates •
 - microservice compatibility •
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AI Requirements .6

Anomaly Detection Models 6.1

:AI models shall detect anomalies in

- financial transactions •
- expense patterns •
- revenue patterns •

:Techniques

- Isolation Forest •
- Neural Networks •

Statistical models •

Document AI 6.2

:The system shall analyze documents including

invoices •

receipts •

financial statements •

:Using

OCR •

NLP •

Risk Scoring 6.3

:The AI system shall calculate

Financial Risk Score

:Based on

transaction anomalies •

compliance violations •

historical patterns •

Future Enhancements .7

:Future system capabilities may include

- predictive financial risk analysis •
 - real-time transaction monitoring •
 - blockchain audit trails •
 - integration with ERP systems •
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Appendix .8

:Example data sources

- Accounting systems •
- Bank statements •
- Invoices •
- Tax records •